Nicolò Iacobone

Computer Vision & 3D Deep Learning Researcher

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Research Profile

Computer vision researcher working across feed-forward 3D reconstruction and semantic scene understanding. Seeking research or ML engineering roles in industry, or a PhD in 3D perception and representation learning.

Research Experience

Research Collaborator — ETH Zürich, Photogrammetry & Remote Sensing (D-BAUG) **Mar 2026 – Sep 2026**
3D-Consistent Instance Segmentation on Frozen Feed-Forward Backbones · *Code*

- Trained a MaskDINO-style decoder on a strictly frozen VGGT-1B backbone for multi-view-consistent 3D instance segmentation, supervised **only by 2D ScanNet v2 annotations**; per-scene feature caching cut training to ∼0.8 GPU-days, against ∼16 for the two published baselines.
- Designed a query set shared across view bundles, so one query tracks one instance across all frames by construction, via cross-frame attention and 3D anchors, the two dominant mechanisms in the ablation.
- Built the training and evaluation stack end to end: four dataset adapters (ScanNet v2/ScanNet++/Replica/Aria Synthetic Environments) and the fully unposed 2D→3D bridge (Sim(3)+ICP, voting-based lifting), scored on the official, unmodified ScanNet evaluator.
- Official ScanNet 3D benchmark, unposed, class-agnostic, 50 views: **0.053 / 0.170 / 0.542** (AP/AP50/AP25), ahead of FAST3DIS (0.038 / 0.096 / 0.316) and IGGT (0.028 / 0.112 / 0.287) on all three; both are zero-shot on ScanNet.
- Data-scaling curves, ablations against a measured seed spread and zero-shot transfer to ScanNet200/ScanNet++/Replica localised the residual loss in 2D→3D lifting, not the decoder.

Master's Thesis — ETH Zürich **Jul 2025 – Mar 2026**
Towards Multi-View 3D Instance Segmentation via Semantic Distillation · *Code* · *SAM 2 teacher fork*

- Designed a **semantic distillation** framework: a parallel DPT head on MapAnything regresses frozen SAM 2 features under a geometry-aware multi-view consistency loss (projection, Z-buffer occlusion, confidence weighting).
- Built the training stack (progressive DINOv2 unfreezing, bf16 AMP distributed on SLURM) and evaluation tooling: sharded COCO teacher-feature extraction, student features decoded by SAM 2's own mask decoder, Rerun PCA views.
- Identified a **capacity bottleneck** of geometric encoders under fine-grained semantic regression, motivating the work above.

Annotation & Synthetic Data Pipelines

Independent work on open-source models **Jul 2025 – Mar 2026**

- **Automatic 2D instance annotation of ScanNet with SAM 3** (code): resumable SLURM pipeline producing per-scene **pseudo-labels**, with persistent masklet tracking, IoU re-association of lost tracks, Florence-2 region proposals in place of hand-written class prompts, and per-scene QA output.
- **Controlled synthetic 3D data and point tracking** (Kubric, SpaTrackerV2): MOVi scenes with controlled camera motion and ground-truth instance masks, covering static, dynamic, occlusion-heavy and many-object cases, with batch 3D point tracking seeded from those masks.

Education

MSc in Computer Engineering — Artificial Intelligence and Data Analytics **Oct 2022 – Mar 2026**
Politecnico di Torino, Italy · 108/110 · Thesis carried out at ETH Zürich (Jul 2025 – Mar 2026)

BSc in Computer Science and Automation Engineering **Oct 2019 – Oct 2022**
Politecnico di Bari, Italy · Erasmus exchange at Universitatea Tehnică din Cluj-Napoca, Romania (Oct 2021 – Jul 2022)

Academic Projects

Real-Time Anomaly Detection for Road Scenes: real-time inference on embedded devices Jun 2024 · Report
Synthetic Data Generation for AROL Closure Systems: CVAE for industrial sensor time series Mar 2024 · Code & slides

Skills

3D Computer Vision: multi-view 3D reconstruction · 2D/3D instance & semantic segmentation · point tracking · feed-forward geometry models (VGGT, MapAnything) · unposed 2D→3D lifting, Sim(3)/ICP alignment

Deep Learning: DETR / Mask2Former / MaskDINO-family architectures · deformable attention · Hungarian matching · distillation & frozen foundation models (SAM 2/3, Florence-2) · synthetic data generation (Kubric) · CVAE

Research Engineering: PyTorch, Hydra, bf16 AMP / DDP · HPC / SLURM, CUDA builds · ScanNet-family benchmarks

Programming: Python (advanced) · C/C++ **Languages:** Italian (native) · English (proficient)